

Ezekiel Talampas

+63 9151607291 | kielctalampas@gmail.com | linkedin.com/in/ezekielctalampas/ | github.com/kieltalampas

EDUCATION

Mapua Malayan Colleges Laguna

2022 – 2026

Bachelor of Science in Computer Science, Specialization in Data Science and Analytics

EXPERIENCE

Software Engineering Intern — STMicroelectronics

May 2026 – Jul 2026

- Designed and built the TNF Reference Portal, a Flask web application consolidating product documentation (overview, process flow, materials, documents, deployment) previously scattered across separate files, with role-based admin access and CRUD functionality for department-wide use
- Automated a manual PPM and Yield reporting process by building four Power BI dashboards across two departments (Finishing and Test), using Power Query for data cleaning and a relational data model integrating over two years of historical production data
- Collaborated directly with engineers and supervisors, most without a software background, to translate operational needs into working systems through iterative requirement gathering and department-wide progress presentations

PROJECTS

MakiPrint

Mapúa Malayan Colleges Laguna

- Built a web-based annotation tool that produced a custom dataset of 1,780 labeled document page images across three ink intensity categories, validated through Krippendorff's Alpha ($\alpha = 0.886$) with Naive Bayes tie resolution
- Trained a ResNet-18 convolutional neural network achieving 95.13% test accuracy on ink classification, with a comparative study showing a 3.74 percentage point improvement when CMYK metadata features were fused with image inputs
- Developed a full-stack web-based printing management platform deployed at a partner retail printing shop, integrating automated ink-based cost estimation, online payment, and administrative order management

WHO Global Disease Outbreak Pipeline

End-to-end AWS data pipeline processing WHO global disease outbreak data with automated scheduling and monitoring for analytics reporting

- Built a PySpark ETL pipeline in AWS Glue processing 3,500+ disease outbreak records across 90+ diseases and 236 countries
- Designed a Redshift data warehouse with optimized schema and views for Power BI analytics reporting
- Implemented automated weekly scheduling with CloudWatch monitoring for pipeline failure alerts

AQI Prediction System

Web-based Air Quality Index classification system built with machine learning and real-world air pollution data to predict AQI severity from atmospheric pollutant inputs

- Trained a Gaussian Naive Bayes classifier to predict Air Quality Index across 6 severity categories
- Applied information gain analysis to rank feature importance of 4 pollutants (PM2.5, O₃, NO₂, CO)
- Built a multi-page interface with real-time prediction, feature analysis, and history tracking

Stock Price Analytics Pipeline

End-to-end analytics pipeline transforming raw stock market data into optimized dimensional models using the medallion architecture pattern for analytics performance

- Built Bronze → Silver → Gold transformations using Delta Live Tables
- Implemented fact and dimension tables for analytics optimization
- Loaded curated data into Snowflake for BI reporting

ACTIVITIES

- Junior Philippine Computer Society (JPCS) — Member
- JPCS Mapúa MCL 2nd AI Summit — attended summit themed “Redefining Intelligence: What it means to be Human in an AI world”
- Introduction to the Data Center Industry — Digital Halo seminar on industry infrastructure supporting SDG 9: Industry, Innovation, and Infrastructure

CERTIFICATIONS

IBM AI Developer, Google Data Analytics, Python for Data Science, AI & Dev, Google Cloud Computing Foundations: Cloud Computing Fundamentals

SKILLS

Languages & Tools: Python, SQL, PySpark, Databricks, Snowflake, AWS, Power BI, Data Analysis, Cloud Computing, Git